

JACC STATE-OF-THE-ART REVIEW

Obesity and Heart Failure With Reduced Ejection Fraction

A Review

Muhammad Shahzeb Khan, MD, MSc,^{a,b,c,*} Syed Sarmad Javaid, MBBS,^{d,*} Mark C. Petrie, MD,^e Shelley Zieroth, MD,^f Stefan D. Anker, MD, PhD,^{g,h} Javed Butler, MD, MPH, MBA^{a,d}

ABSTRACT

Obesity and heart failure with reduced ejection fraction (HFrEF) frequently coexist. In fact, most patients with HFrEF are overweight or have concomitant obesity. However, despite this high prevalence, obesity in HFrEF has received limited attention as most recent reviews have concentrated on heart failure broadly or on the heart failure with preserved ejection fraction (HFpEF) phenotype. This limited attention may, in part, be attributed to the stronger and more consistent pathophysiological and epidemiologic associations observed between obesity and HFpEF, whereas the relationship between obesity and HFrEF remains less well characterized. Much of this uncertainty reflects inconsistent findings across studies regarding whether body mass index independently predicts incident HFrEF, given its well-recognized limitations as a surrogate marker of adiposity. These challenges are further compounded by the phenomenon known as the obesity paradox in patients with HFrEF, in which higher body mass index is often associated with better outcomes. Nonetheless, in HFrEF, obesity may be better conceptualized as a modifiable disease amplifier that influences disease progression and clinical trajectory, rather than a primary etiologic driver. Weight loss interventions such as obesity pharmacotherapy have demonstrated safety and efficacy in patients with obesity and HFpEF, where obesity-related pathophysiologic processes including systemic inflammation, neurohormonal activation, local paracrine signaling, and mechanical loading are well-described contributors to disease development. Although the relative contribution of these processes may differ in HFrEF, excess adiposity may still exacerbate disease progression through these pathways, suggesting potential applicability of weight loss interventions in this population. To date, evidence in HFrEF remains limited and is derived largely from non-obesity-focused trials with methodological constraints. Despite the development of several novel weight loss agents, none are being specifically evaluated in patients with HFrEF, limiting understanding of their potential clinical impact in these patients. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

From the ^aBaylor Scott and White Research Institute, Dallas, Texas, USA; ^bDepartment of Medicine, Baylor School of Medicine, Temple, Texas, USA; ^cBaylor Scott and White Health: The Heart Hospital Plano, Plano, Texas, USA; ^dDepartment of Medicine, University of Mississippi Medical Center, Jackson, Mississippi, USA; ^eBritish Heart Foundation Cardiovascular Research Centre, University of Glasgow, Glasgow, United Kingdom; ^fSection of Cardiology, Max Rady College of Medicine, University of Manitoba, Winnipeg, Manitoba, Canada; ^gDepartment of Cardiology (CVK) of German Heart Center Charité; German Centre for Cardiovascular Research (DZHK) partner site Berlin, Charité Universitätsmedizin, Berlin, Germany; and the ^hInstitute of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland. *These authors contributed equally to this work as first authors.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received November 13, 2025; revised manuscript received February 2, 2026, accepted February 9, 2026.

**ABBREVIATIONS
AND ACRONYMS****BMI** = body mass index**cAMP** = cyclic adenosine
monophosphate**EAT** = epicardial adipose tissue**EF** = ejection fraction**FDA** = U.S. Food and Drug
Administration**GDMT** = guideline-directed
medical therapy**GLP-1 RA** = glucagon-like
peptide-1 receptor agonist**HF** = heart failure**HFpEF** = heart failure with
preserved ejection fraction**HFrEF** = heart failure with
reduced ejection fraction**LV** = left ventricular**LVAD** = left ventricular assist
device**LVEF** = left ventricular
ejection fraction**NT-proBNP** = N-terminal
pro-B-type natriuretic peptide**T2DM** = type 2 diabetes
mellitus

Obesity is a common comorbidity in heart failure (HF), affecting nearly one-half of all patients.¹ Although the contribution of obesity to heart failure with preserved ejection fraction (HFpEF) is well established,² its role in heart failure with reduced ejection fraction (HFrEF) remains less clearly defined. Contemporary HFrEF populations are characterized by a high burden of overweight and obesity, with most participants in major clinical trials and registries falling within overweight or obese body mass index (BMI) ranges.³⁻⁵ This consistent pattern underscores that excess adiposity is a pervasive clinical context in HFrEF rather than an incidental finding. However, despite this high prevalence, obesity in HFrEF has received comparatively limited attention, as recent literature has largely focused on HF broadly or on the HFpEF phenotype.^{6,7}

This limited focus may partly reflect the historical design of foundational HFrEF trials, in which excessive adiposity was not regarded as a key determinant of disease progression and BMI was frequently omitted as a baseline variable.^{8,9} In addition, epidemiologic associations between adiposity and HF risk appear stronger and more consistent for HFpEF than for HFrEF, contributing to uncertainty regarding the etiologic role of obesity in HFrEF.¹⁰⁻¹⁵ Reliance on BMI as a surrogate measure of adiposity has further complicated interpretation of these relationships. Taken together, these observations highlight heterogeneity in the available evidence and support a cautious interpretation of obesity as a primary causal factor in HFrEF.

Nevertheless, irrespective of causality, obesity in HFrEF may be better conceptualized as a modifiable disease amplifier that shapes disease progression and clinical trajectory, rather than a primary etiologic driver. Obesity is characterized by activation of the renin-angiotensin-aldosterone system,⁸ suggesting that excess adiposity may intensify neurohormonal stress and contribute to worsening disease when both conditions coexist. The clinical implications of obesity extend beyond HF hospitalization risk, as severe obesity, defined as BMI ≥ 35 kg/m², can limit eligibility for heart transplantation, thereby narrowing treatment options for individuals with advanced disease.¹⁶ Overall, these observations indicate that obesity can shape the clinical course and influence therapeutic decisions across the HFrEF spectrum. Given that obesity-directed therapies have the

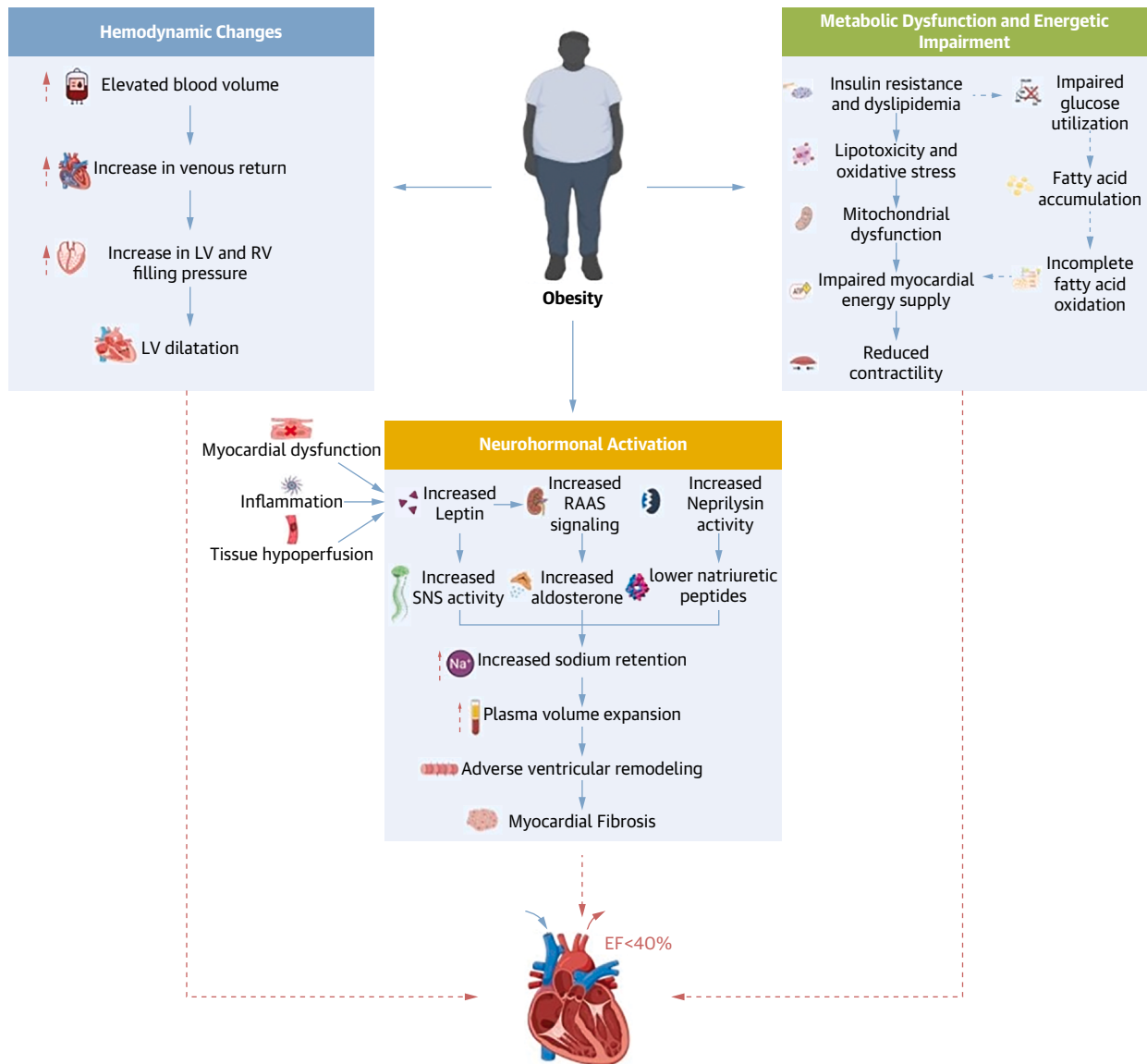
potential to improve symptoms, functional capacity, quality of life, and metabolic health,⁶ incorporating these approaches into HFrEF care warrants consideration.

Despite these potential therapeutic implications, prior reviews have not provided a focused evaluation of obesity in the context of HFrEF.^{6,7} Recent broad syntheses of obesity and HF have highlighted the relevance of adiposity across HF phenotypes but have largely adopted a generalized perspective, with limited attention to the distinct biological and clinical features of HFrEF.⁸ Consequently, obesity-related mechanisms and therapeutic considerations specific to HFrEF remain incompletely defined. In this review, we conceptualize obesity in HFrEF as a modifiable risk factor that amplifies disease progression rather than a primary etiologic driver. We delineate obesity-related biological pathways in HFrEF and contrast their relative importance with those described in HFpEF, where obesity often plays a more direct etiologic role. We integrate trial-level evidence on incretin-based therapies and provide a mechanistic rationale for evaluating these agents in HFrEF. We also discuss other obesity management strategies in HFrEF, including lifestyle interventions and bariatric surgery, and outline key clinical and methodological considerations to inform future trial design and therapeutic development in this population.

**PATHOPHYSIOLOGICAL MECHANISMS BY
WHICH OBESITY WORSENS HFrEF**

Obesity is a well-established contributor to HFpEF.¹⁷ In contrast, its role in HFrEF differs, with obesity functioning less as a primary etiologic driver and more as a disease modifier that amplifies underlying pathophysiologic processes by increasing neurohormonal activation, hemodynamic load, and metabolic stress in hearts already affected by systolic dysfunction.^{8,18} Obesity commonly coexists with cardiometabolic conditions including diabetes, hypertension, and coronary artery disease, all of which are independent risk factors for the development of HFrEF (Figure 1).¹⁹⁻²¹

NEUROHORMONAL ACTIVATION. Neurohormonal activation is a fundamental hallmark of HFrEF and triggers chronic sympathetic nervous system stimulation and renin-angiotensin-aldosterone system upregulation.²² Sustained activation leads to vasoconstriction, sodium and water retention, adverse ventricular remodeling, and progressive systolic dysfunction. Obesity amplifies this neurohormonal state. Several analyses have shown that patients with

FIGURE 1 Pathophysiological Mechanisms By Which Obesity Worsens HFrEF

This figure illustrates the proposed mechanisms through which obesity exacerbates HFrEF progression, including metabolic dysfunction, hemodynamic changes, and neurohormonal activation. Arrows indicate the directional flow of pathophysiologic processes. HFrEF = heart failure with reduced ejection fraction; LV = left ventricle; RAAS = renin-angiotensin-aldosterone system; RV = right ventricle; SNS = sympathetic nervous system.

higher BMI have elevated circulating aldosterone concentrations and lower B-type natriuretic peptide/ N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels,²³ suggesting potentiation of the renin-angiotensin-aldosterone system and augmented neprilysin activity. Packer's leptin-aldosterone-neprilysin framework provides further mechanistic insight into this interaction.^{18,24}

In HFrEF, leptin levels rise primarily in response to myocardial dysfunction, inflammation, and tissue hypoperfusion rather than adiposity, which differs from HFpEF in which obesity is the main driver of circulating leptin.¹⁸ Despite these differing sources, elevated leptin in HFrEF enhances sympathetic and renin-angiotensin-aldosterone system activation, and contributes to profibrotic signaling.¹⁸

HEMODYNAMIC LOAD ALTERATIONS. Obesity also induces hemodynamic alterations characterized by increased total blood volume and elevated venous return,²⁵ driven by higher metabolic demand and greater tissue mass. In a normally functioning heart, these changes lead to increased cardiac output and, in severe cases, may result in high-output HF.²⁶ However, in patients with HFrEF, the same increase in volume contributes to elevated left- and right-sided filling pressures,²⁷ which in turn promote left ventricular (LV) dilation and hypertrophy. Over time, these adaptations can progress to LV systolic dysfunction due to increased wall stress and myocardial oxygen demand. Neurohormonal activation further amplifies these maladaptive changes by promoting fibrosis and additional fluid retention.

SYSTEMIC INFLAMMATION. Adipose tissue releases proinflammatory cytokines, creating a chronic low-grade inflammatory state that contributes to oxidative stress and myocardial injury.²⁸ However, the role of obesity-related inflammation in HFrEF is less well characterized than in HFpEF. In HFpEF, dysfunctional adipose tissue is a central driver of systemic inflammation, whereas in HFrEF, inflammatory activity more often reflects fluid retention, clinical congestion, or acute decompensation rather than adiposity itself.⁸ Differences in epicardial adipose tissue (EAT) further reinforce this distinction.⁸ EAT is typically expanded in HFpEF but reduced in HFrEF, and lower EAT volumes in HFrEF are associated with worse outcomes,²⁹ likely due to catabolism and cachexia. The EMPA-TROPISM (Are the ‘Cardiac Benefits’ of Empagliflozin Independent of Its Hypoglycemic Activity?) trial adds more nuance to this observation by showing that empagliflozin lowers EAT volume and body weight, causes reverse myocardial remodeling, and improves inflammatory biomarkers.³⁰ This finding suggests that the interpretation of EAT reduction depends on whether it occurs through disease-related wasting or therapeutic modulation. Similarly, adiponectin biology also differs in HFrEF.¹⁸ Generally, obesity is associated with reduced adiponectin levels, whereas in HFrEF, circulating adiponectin levels are paradoxically elevated.¹⁸ This increase likely reflects myocardial adiponectin release as systolic function declines.^{18,31} The adiponectin levels may decrease with effective therapy; however, rising levels despite treatment suggest ongoing myocardial stress and are associated with higher mortality.¹⁸ Altogether, these findings indicate that adipokine and inflammatory profiles in HFrEF differ from those seen in obesity-driven HFpEF and may follow counterintuitive patterns.

METABOLIC DYSFUNCTION AND CARDIAC ENERGETICS.

Obesity also impairs cardiac function through metabolic alterations such as insulin resistance and dyslipidemia.³² These conditions promote cellular injury through mechanisms like lipotoxicity and oxidative stress, ultimately leading to mitochondrial dysfunction and reduced cardiomyocyte contractility.³³ One key consequence of these changes is a disruption in the heart’s normal energy metabolism. In HF, the myocardium typically attempts to conserve energy by shifting from fatty acid oxidation to more efficient glucose utilization.³⁴ However, in individuals with obesity, insulin resistance impairs this adaptive response, preventing the heart from effectively switching to glucose as a fuel source.³⁵ As a result, the myocardium becomes overloaded with fatty acids it cannot efficiently metabolize, creating an energy deficit that becomes especially detrimental during periods of increased demand. This imbalance is reflected by an increased resting ATP demand and reduced ATP availability during exertion.³⁶ In cases of severe obesity, mitochondrial uncoupling further worsens energy inefficiency, compounding the metabolic stress on the failing heart.³⁷

MECHANISTIC AND TRANSLATIONAL RATIONALE FOR INCRETIN-BASED THERAPIES IN HFrEF WITH OBESITY.

Incretin-based therapies have the potential to modulate multiple pathophysiological pathways through which obesity influences HFrEF progression. Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) and dual incretin agonists improve insulin resistance, glycemic control, and lipid profiles,³⁸⁻⁴⁰ supporting their ability to address metabolic abnormalities that frequently accompany HFrEF in the setting of obesity. Beyond metabolic effects, GLP-1 RAs also may influence neurohormonal activity. Experimental and clinical studies indicate attenuation of renin-angiotensin-aldosterone system signaling through reduced angiotensin II formation, enhanced degradation, promotion of natriuresis via sodium-hydrogen exchanger 3 inhibition, and acute reductions in aldosterone levels.⁴¹ Collectively, these effects support more favorable volume regulation and reduced cardiac loading conditions, which are central to HFrEF pathophysiology. Importantly, GLP-1 RAs are associated with modest reductions in systolic blood pressure in non-HFrEF populations (approximately 2-5 mm Hg),⁴² indicating a favorable hemodynamic profile that may allow concomitant use with guideline-directed medical therapy (GDMT), particularly in patients whose ability to tolerate GDMT is constrained by low blood pressure.⁴³ However, the blood pressure effects of GLP-1 RAs and

other weight loss drugs in HFrEF remain uncertain and will become evident in large, dedicated randomized controlled trials. In addition, the anti-inflammatory effects of GLP-1 RAs also may help mitigate the low-grade systemic inflammation commonly observed in patients with HFrEF and obesity.⁴⁴ Furthermore, emerging data suggest that GLP-1 RAs are well tolerated and associated with clinically meaningful weight loss in patients with left ventricular assist devices (LVADs).⁴⁵ Given that individuals with class II or III obesity supported with LVADs undergo heart transplantation at significantly lower rates (approximately 25%-57%) than leaner counterparts due to inadequate weight loss,^{46,47} the sustained weight reduction achieved with GLP-1-based therapies may offer a pathway for selected patients to attain the BMI reductions necessary to favorably influence transplant candidacy. Nonetheless, the safety and efficacy of GLP-1 RAs are currently unknown in all HFrEF populations, including those with advanced HFrEF.

OBESITY PARADOX IN HFrEF

To explain the obesity paradox in individuals with HFrEF, several theories have been proposed. One hypothesis suggests that the greater metabolic reserves and muscle mass in individuals with obesity provide a protective buffer against the catabolic stress of advanced HF.⁴⁸ Another theory suggests that the increased blood volume and higher baseline blood pressure observed in obesity may improve tolerance to cardioprotective medications.⁴⁹ Furthermore, some evidence indicates that left ventricular ejection fraction (LVEF), a key prognostic indicator, improves more frequently in patients with obesity, potentially contributing to their enhanced survival.⁴⁹

Despite these proposed mechanisms, the obesity paradox has been questioned on methodological grounds, particularly due to the widespread reliance on BMI as a surrogate for adiposity. Most supporting studies use BMI, which does not distinguish between fat and lean mass or adequately capture fat distribution and fluid retention.⁵⁰⁻⁵³ These limitations are especially relevant in HFrEF, in which volume overload can inflate body weight and BMI, thereby obscuring the true relationship between adiposity and clinical outcomes.⁵⁴ Because BMI incompletely reflects body composition, the apparent survival advantage associated with higher BMI may reflect measurement bias rather than a protective biological effect. Consistent with this interpretation, studies using alternative anthropometric measures of fat distribution have identified a different risk profile.

In a large cohort of patients with HFrEF, adjustment for confounders attenuated the survival advantage associated with higher BMI, whereas greater central adiposity, assessed by waist-to-hip ratio, was associated with an increased risk of cardiovascular death and HF hospitalization.⁵⁵ Similar patterns have been observed in nonischemic HFrEF populations.⁵⁶ Nevertheless, some studies have reported improved outcomes with greater central adiposity,⁵⁷ underscoring heterogeneity in the evidence and suggesting that prognostic effects of adiposity extend beyond simple anthropometric measures.

Last, changes in body composition, particularly weight loss, further complicate any interpretation of the obesity paradox.⁴⁹ Observational studies have consistently linked weight loss to increased mortality, regardless of a patient's baseline BMI.⁵⁸ However, this association is often confounded by the critical difference between intentional weight loss and unintentional weight loss due to cardiac cachexia, a catabolic state indicating advanced disease and poor prognosis.⁴⁹ In HFrEF, disease progression often induces the loss of lean muscle mass, leading to reductions in BMI that reflect muscle wasting rather than healthy fat loss. Consequently, studies that fail to distinguish between these 2 types of weight loss may erroneously conclude a survival advantage for patients who remain obese,⁵⁹ underscoring a critical need for randomized controlled trials to evaluate the true impact of intentional weight loss in HFrEF.

WEIGHT LOSS INTERVENTIONS IN HFrEF

Weight loss interventions including lifestyle modification, weight loss medications, and bariatric surgery, have demonstrated potential benefits in HF.⁶⁰ However, evidence specific to the HFrEF phenotype remains limited and inconclusive. In the following sections, we summarize the current data on these interventions in patients with HFrEF to provide a clearer understanding of the evidence landscape for weight loss strategies in this subgroup ([Table 1](#)).

WEIGHT LOSS MEDICATIONS. GLP-1 agonists.

Currently, evidence regarding weight loss medications in HFrEF primarily focuses on GLP-1 RAs, stemming from various trials and secondary analyses.⁶¹⁻⁶⁹ In the FIGHT (Heart Failure Network Functional Impact of GLP-1 for Heart Failure Treatment) trial, 300 patients with advanced HFrEF were randomized to receive either liraglutide (n = 154) or placebo (n = 146). The population included in this trial was not an obese HFrEF population, and the dose of liraglutide (1.8 mg)

used was not a weight loss dose. Patients had severe HFrEF, with all having been hospitalized within the previous 2 weeks, and 80% having experienced prior hospitalization within the preceding 12 months. The mean LVEF was 25%, and the mean NT-proBNP was 2,000 pg/mL. At 6-month follow-up, there was no difference in the primary endpoint, which included a global rank score based on time to death, time to HF rehospitalization, and time-averaged change in NT-proBNP.⁶² Liraglutide also failed to show favorable effects on secondary endpoints such as echocardiographic parameters and patient-reported health status. A post hoc analysis of this small trial suggested a numerically higher, although nonsignificant, risk of total HF hospitalizations or all-cause death with liraglutide, which appeared more pronounced in patients with NYHA functional class III-IV HF.⁶³

Similarly, a post hoc analysis of the EXSCEL (Exenatide Study of Cardiovascular Event Lowering) trial ($n = 455$) indicated a potential association between exenatide and an increased risk of HF hospitalization in patients with LVEF $<40\%$.⁶⁴ However, this was not an obesity-focused trial, and the exenatide dose (2 mg) used was not weight loss GLP-1 RA preparation. The study also had several important limitations, including the lack of a standardized approach for characterizing HF, reliance on investigator-classified LVEF categories without precise ejection fraction (EF) measurements, and a lack of systematic data on symptomatic status, natriuretic peptide levels, and prior hospitalizations. Moreover, only 10% of the participants had LVEF $<40\%$ at baseline. Although these limitations warrant major caution in interpreting the EXSCEL findings, concerns regarding an increased risk of HF hospitalization with GLP-1 RAs were suggested in the combined analysis of the EXSCEL and FIGHT trials.⁶⁵

The LIVE (Effect of Liraglutide on Left Ventricular Function in Stable Chronic Heart Failure Patients with and without Diabetes) trial further explored liraglutide's role in 241 patients with stable HFrEF both with and without type 2 diabetes mellitus (T2DM) who were receiving optimal HF therapy.⁶¹ Again, the population studied was not an obese HFrEF population and the dose of liraglutide (1.8 mg) was not a weight loss dose. At 24 weeks, liraglutide did not change LVEF. A higher rate of "serious cardiovascular events" was reported but there was only 1 death and 1 HF hospitalization in the whole trial. Another study evaluating a GLP-1 RA (albiglutide 30 mg) for myocardial metabolic abnormalities in chronic HF (LVEF $<40\%$) found no effects on cardiac function or myocardial glucose utilization after

12 weeks, although a significant increase in peak oxygen consumption (VO_2) was observed and attributed to noncardiac mechanisms.⁶⁶ In contrast, a small trial that enrolled 52 patients with HF and moderately reduced EF (LVEF $<50\%$) found that 1 week of liraglutide treatment was associated with favorable changes in markers of inflammation, oxidative stress, and LV function.⁶⁷

Recently, the SELECT (Semaglutide Effects on Cardiovascular Outcomes in People with Overweight or Obesity) trial demonstrated the superiority of semaglutide over placebo in reducing cardiovascular events among patients with obesity and preexisting cardiovascular disease.⁶⁸ Within this trial, 4,286 participants (24%) were reported by investigators to have HF at baseline, including 1,341 labeled as having HFrEF, making it the largest GLP-1 RA study with HFrEF.⁶⁸ However, there was no standardized method for characterizing HF: details regarding symptomatic status, EF measurement, thresholds defining HFrEF, natriuretic peptide levels, and prior HF hospitalizations were not systematically recorded.

A recent report from the SELECT trial, which stratified patients by HF subtype, showed that semaglutide was associated with a reduction in major adverse cardiac events in both HFpEF (HR: 0.65; 95% CI: 0.49-0.87) and HFrEF (HR: 0.69; 95% CI: 0.51-0.91), with no indication of treatment heterogeneity between the groups.⁶⁹ Similar trends were observed across other outcomes, including HF composite, cardiovascular death, and all-cause death. These findings suggest that semaglutide may provide clinical benefit in patients with HFrEF, similar to those seen in HFpEF. Notably, this stands in contrast to earlier, smaller studies such as LIVE and FIGHT, which included patients without obesity and were not adequately powered to assess clinical outcomes.

Overall, current data do not demonstrate benefit of GLP-1 RAs in patients with HFrEF, which stands in contrast to the improvements reported in obesity-related HFpEF. However, variations in trial methodology limit the ability to compare these findings directly. For instance, the FIGHT trial enrolled a high-risk HFrEF cohort with recent decompensation requiring hospitalization within the preceding 2 weeks, whereas the HFpEF trials excluded individuals hospitalized within 30 days to yield a more clinically stable population.^{70,71} A further distinction is that in existing HFrEF studies, obesity was not an inclusion criterion, which led to the use of non-weight loss doses that achieved only modest reductions in body weight. Follow-up duration also differed because HFpEF trials extended only 52 to 104 weeks, whereas most HFrEF trials were shorter

TABLE 1 Summary of Studies Evaluating Weight Loss Strategies in Patients With HFrEF

Author, Year	Study Design	Study Objectives	Sample Size	Inclusion Criteria	BMI, kg/m ²	LVEF, %	Follow-up, wk	Comparison	Outcomes
Exercise intervention									
Vetrovsky et al, 2024	Randomized clinical trial	To evaluate whether lifestyle walking intervention with self-monitoring and regular phone counseling improves functional capacity in patients with stable HFrEF	202	Stable HFrEF (LVEF <40%) with NYHA functional class II-III symptoms on evidence-based medical therapy at maximally tolerated doses	29 ^a	32.5 ^a	26 ^b	Physical activity vs usual care	Change in 6MWT (m): 35.2 vs 26.9, <i>P</i> > 0.05
Feuerstein et al, 2023	Randomized clinical trial	To assess the effect of supervised exercise training on exercise capacity in patients with advanced HF and LVADs	64	Advanced CHF on guideline-directed medical therapy, supported with an LVAD for ≥3 months	29.9 ± 5.4	—	24	Exercise training vs usual care	Change in 6MWT (m): 22.5 vs −15.7, <i>P</i> = 0.06 Change in peak VO ₂ (mL/kg/min): 0.42 vs −0.47, <i>P</i> = 0.08
Lan et al, 2020	Randomized clinical trial	To evaluate the impact of different exercise modalities on cardiac function and cardiorespiratory fitness	36	HFrEF (EF <50%) and receiving stable medical therapy ≥3 months	29.6 ± 1.1	31 ± 3	12	Aerobic exercise vs control	Change in peak VO ₂ (mL/kg/min): 2.7, <i>P</i> < 0.001 vs −1, <i>P</i> < 0.01 Change in LVEF (%): 3, <i>P</i> > 0.05 vs 1, <i>P</i> > 0.05 Change in exercise duration (s): 127, <i>P</i> < 0.001 vs −27, <i>P</i> > 0.05
								Resistance exercise vs control	Change in peak VO ₂ (mL/kg/min): 2.7, <i>P</i> < 0.001 vs −1, <i>P</i> < 0.01 Change in LVEF (%): 3, <i>P</i> < 0.05 vs 1, <i>P</i> > 0.05 Change in exercise duration (s): 93, <i>P</i> < 0.05 vs −27, <i>P</i> > 0.05
Groennebaek et al, 2019	Randomized clinical trial	To investigate whether BFRRE could improve functional capacity and QoL in patients with CHF	36	LVEF ≤45% and NYHA functional class I-III	28.1 ± 4.2	35.7 ± 6.7	6	BFREE (resistance exercise) vs control	Change in 6MWT (m): 39.0, <i>P</i> = 0.019 Change in QoL (MLHFQ score): 5.4, <i>P</i> = 0.052
de Meirelles et al, 2014	Randomized clinical trial	To evaluate the effect of exercise training on cardiorespiratory fitness and inflammatory biomarkers in patients with HF	30	Physician-diagnosed HF (LVEF ≤40%), exercise capacity ≥3 METs, clinically stable ≥3 months	28.3 ± 0.8	31.5 ± 1.8	26 ^b	Exercise training vs control	Change in peak VO ₂ (mL/kg/min): 5.8 vs 0.1, <i>P</i> < 0.05 Change in CRP levels (mg/dL): −0.17 vs 0.02, <i>P</i> < 0.05
Horwich et al, 2012	Secondary analysis of HF-ACTION Trial	To evaluate whether overweight and obese patients with HF derive different benefits from exercise training with respect to health status and clinical outcomes	2,314	LVEF ≤35%, NYHA functional class II-IV, stable on optimal medical therapy for ≥6 weeks, and able to exercise	—	25 ^a	12	Exercise training vs usual care	QoL improved across range of BMI categories with exercise training
Gademan et al, 2008	Non-randomized study	To evaluate the effect of exercise training on cardiorespiratory fitness in patients with CHF	34	NYHA functional class II-III, systolic HF, LVEF <45%	28 ± 3.3	34 ± 6	—	Exercise training vs control	Change in peak VO ₂ (%): 14 vs −1, <i>P</i> < 0.01 Change in oxygen uptake efficiency (%): 19 vs 5, <i>P</i> = 0.001

Continued on the next page

TABLE 1 Continued

Author, Year	Study Design	Study Objectives	Sample Size	Inclusion Criteria	BMI, kg/m ²	LVEF, %	Follow-up, wk	Comparison	Outcomes
Evangelista et al, 2006	Randomized clinical trial	To evaluate the impact of a home-based exercise program on weight loss, functional status, psychological well-being, and rehospitalization in overweight and obese patients with advanced HF	99	Advanced HF (LVEF ≤40%), NYHA functional class II-IV	30.5 ± 4.2	27 ± 6.5	26 ^b	Exercise training vs control	Change in BMI (kg/m ²): -2.51 vs 0.5, <i>P</i> = 0.002 Change in 6MWT (m): 198 vs -43, <i>P</i> = 0.514
Corvera-Tindel et al, 2004	Randomized clinical trial	To evaluate the effects of a home walking exercise program on functional status and symptoms in patients with HF	79	NYHA functional class II-IV, LVEF ≤40%, HF duration ≥3 months	29.5 ± 6.3	26.9 ± 8.7	12	Exercise training vs usual activity	Change in 6MWT (feet): 118.1 vs -9.3, <i>P</i> = 0.001 Post-global rating of symptoms: 3.2 vs 3.7, <i>P</i> = 0.03
Cardiac rehabilitation									
Piotrowicz et al, 2020	Randomized clinical trial	To evaluate whether improvements in QoL achieved through an HCTR program in HF patients are associated with better clinical outcomes	850	NYHA functional class I-III HF and LVEF ≤40%, hospitalized for a cardiovascular event within 6 months prior to randomization	28.9 ± 4.9	30.5 ± 7	9	HCTR vs usual care	Change in 6MWT (m): 30 vs 20.7, <i>P</i> = 0.01 Change in cardiopulmonary exercise time (s): 45.5 vs 16.7, <i>P</i> < 0.001 Change in peak Vo ₂ (mL/kg/min): 0.95 vs 0.00, <i>P</i> < 0.001 Change in QoL (SF-36): 1.58 vs 0.00, <i>P</i> = 0.008
Dalal et al, 2019	Randomized clinical trial	To evaluate the effects of a home-based rehabilitation program on QoL in patients with HFrEF	216	HFrEF (LVEF <45%) confirmed by echocardiography or angiography within the past 5 years	28 ^c	34 ^a	52 ^b	Home-based cardiac rehabilitation vs usual care	Difference in QoL (MLHFQ score overall): -5.7, <i>P</i> = 0.025 Overall hospitalization: 19 vs 24, <i>P</i> = 0.386
Kerrigan et al, 2014	Randomized clinical trial	To examine the effects of cardiac rehabilitation on functional capacity and health status in patients with newly implanted LVAD	26	Recently implanted continuous-flow LVAD, without major comorbidities or limitations that could interfere with exercise training	27 ± 4.5	21 ± 8	6	Cardiac rehabilitation vs usual care	Change in treadmill time (min): 3.1 vs 0.8, <i>P</i> = 0.001 Change in Vo ₂ (L/min) : 0.14 vs 0.08, <i>P</i> = 0.270 Change in 6MWT (m): 52.3 vs 19.4, <i>P</i> = 0.241
Dietary interventions									
Pritchett et al, 2012	Pilot randomized clinical trial	To evaluate the feasibility, safety, and impact of a lifestyle modification program on intentional weight loss in patients with systolic HF	20	Systolic HF (EF ≤50%), NYHA functional class II-III, and metabolic syndrome	39 ± 5.5	26 ± 12	13 ^b	Lifestyle modification (diet and exercise) vs control	Change in weight (kg): -1.2 vs -0.56, <i>P</i> = 0.71 Change in waist circumference (in): 1.8 vs -0.7, <i>P</i> = 0.05 Change in glucose levels (mg/dL): -16.4 vs 2.8, <i>P</i> = 0.37 Change in TG levels (mg/dL): 11.8 vs 5.7, <i>P</i> = 0.87 Change in 6MWT (feet): 169 vs 84, <i>P</i> = 0.59 Change in KCCQ score: 6.1 vs 1.7, <i>P</i> = 0.55

Continued on the next page

TABLE 1 Continued

Author, Year	Study Design	Study Objectives	Sample Size	Inclusion Criteria	BMI, kg/m ²	LVEF, %	Follow-up, wk	Comparison	Outcomes
Evangelista et al, 2009	Randomized study	To evaluate the impact of protein-based dietary interventions on cardiometabolic health and QoL	14	BMI ≥ 27 kg/m ² , NYHA functional class II-III HF, and non-insulin-treated type 2 diabetes mellitus	38 $\pm$ 3.6	26 $\pm$ 7.3	12	High-protein diet vs control	Change in weight (kg): -9.9 vs -1.5, $P < 0.05$ Change in total cholesterol (mg/dL): -35.0 vs 16.5, $P < 0.05$ Change in 6MWT (feet): 287.3 vs -138.4, $P < 0.05$ Change in peak $\dot{V}O_2$ (mL/kg/min): 3.1 vs -0.3, $P < 0.05$

^aMedian value. ^bConverted from months to weeks. ^cMedian value in intervention group.

6MWT = 6-minute walk test; BMI = body mass index; BFRRE = blood flow restriction resistance exercise; CHF = chronic heart failure; CRP = C-reactive protein; EF = ejection fraction; HF = heart failure; HCTR = hybrid comprehensive telerehabilitation; HFrEF = heart failure with reduced ejection fraction; KCCQ = Kansas City Cardiomyopathy Questionnaire; LVAD = left ventricular assist device; LVEF = left ventricular ejection fraction; METs = metabolic equivalents; MLHFQ = Minnesota Living with Heart Failure Questionnaire; QoL = quality of life; SF-36 = Short Form-36 Health Survey; TG = triglycerides; $\dot{V}O_2$ = oxygen consumption.

and thus unlikely to capture the metabolic or structural adaptations that generally require longer treatment exposure.⁷⁰⁻⁷² In addition, HFrEF cohorts were often small, heterogeneous, and often inadequately characterized regarding obesity and EF status. Taken together, these methodological limitations suggest that the absence of benefit in current trials should not be interpreted as evidence that GLP-1 RAs are ineffective in this population. Rather, they highlight the need for large, adequately powered and longer trials using contemporary weight loss doses in well-defined HFrEF and obesity cohorts. Last, a better understanding of how obesity might or might not exacerbate HFrEF pathophysiology may help identify the patients most likely to benefit (or possibly be harmed) from GLP-1-based therapies.

Selective off-label use of GLP-1 RAs in HFrEF. At present, routine use of GLP-1 RAs in patients with HFrEF is not recommended. However, in carefully selected clinical scenarios, off-label use may be considered on an individualized basis following shared decision making. Such situations may include patients with HFrEF and severe obesity in whom excess adiposity limits eligibility for heart transplantation, as well as selected individuals supported with durable LVADs who require weight reduction to facilitate transplant candidacy. Available observational data in LVAD-supported patients suggest that GLP-1 RA use has been generally well tolerated and associated with meaningful weight loss, with some individuals achieving reductions sufficient to meet transplant criteria.⁴⁵ However, weight loss in this setting has been accompanied by LVAD speed adjustments,

warranting close clinical monitoring and echocardiographic reassessment during periods of significant weight change. Off-label use also may be considered in stable patients with HFrEF and coexisting obesity-related metabolic disease, such as T2DM or marked insulin resistance, in which GLP-1 RAs may be prescribed primarily for metabolic risk reduction, with attention to preserving lean mass and monitoring hemodynamic tolerance. Conversely, GLP-1 RAs should be avoided in patients with advanced or unstable HFrEF, including those with recent decompensation, cardiac cachexia, unintentional weight loss, or heightened susceptibility to arrhythmias, until more definitive evidence becomes available.

Potential safety concerns with GLP-1 RAs in HFrEF. In patients with HFrEF, reducing heart rate with treatments such as beta-blockers or ivabradine has been associated with improved outcomes.^{73,74} However, weight loss medications such as GLP-1 RAs have been associated with modest increases in heart rate due to their positive chronotropic effects, raising potential concerns about their use in this population.⁷⁵ These effects have been linked to underlying mechanisms involving activation of the cyclic adenosine monophosphate (cAMP)-dependent pathway. GLP-1 receptors, located in cardiomyocytes and sinoatrial node cells, stimulate this pathway, leading to increased intracellular cAMP levels in the myocardium.^{76,77}

In the setting of HF, disruption of regulatory complexes can prolong cAMP activation, resulting in calcium overload and a heightened risk of myocardial dysfunction and arrhythmias.⁶³ Clinical evidence

supports this concern, as agents that elevate cAMP, such as milrinone, have been shown to increase the risk of arrhythmic events and mortality in HFrEF, whereas beta-blockers, which reduce cAMP levels, confer a protective effect.⁷⁸ Because GLP-1 RAs elevate cAMP independently of beta-adrenergic receptors, there is concern that beta-blockers may not fully counteract their potential adverse effects.

Although GLP-1 RAs may pose minimal risk in individuals without myocardial dysfunction or elevated arrhythmia risk, patients with HFrEF characterized by reduced cardiac reserve and increased susceptibility to arrhythmias may be more vulnerable.⁶³ Furthermore, GLP-1 RA-induced weight loss could be harmful in those with advanced HF and cachexia. Given these uncertainties and the complex pharmacodynamic profile of GLP-1 RAs, dedicated randomized trials in patients with T2DM and stage C HF are essential to determine their safety and efficacy in this high-risk population.

Other weight loss medications. Among other medications approved by the U.S. Food and Drug Administration (FDA) for obesity, orlistat is the only pharmacotherapy that has been evaluated in patients with HFrEF, although the evidence is limited to a single small pilot trial. In this study, participants were randomized to receive either orlistat plus standard care ($n = 11$) or standard care alone ($n = 10$) for 12 weeks.⁷⁹ At follow-up, the orlistat group showed significant weight loss and improvements in metabolic parameters, including total cholesterol, low-density lipoprotein cholesterol, and triglycerides. Although improvements in 6-minute walk test distance and NYHA functional class also were observed, these changes were not significant. Importantly, orlistat was well tolerated, with no major safety concerns reported. Despite these encouraging findings, larger trials have not been conducted likely because of concerns about potential drug-drug interactions with medications commonly used in patients with HF, such as amiodarone, which may limit its broader applicability in this population.⁷

LIFESTYLE INTERVENTIONS. Lifestyle interventions such as diet and exercise are key components of obesity management⁸⁰; however, evidence supporting their effectiveness in HFrEF remains limited.⁸¹⁻⁹⁴ Exercise training studies in HFrEF have demonstrated improvements in quality of life, functional capacity, and cardiorespiratory fitness, with some benefits also extending to patients supported with LVADs.⁸²⁻⁸⁹ However, the magnitude and consistency of these effects have varied, likely reflecting differences in patient characteristics, follow-up

duration, and exercise protocols.⁹⁰⁻⁹² Evidence for dietary interventions in patients with obesity and HFrEF is even more limited, as available studies are few and small, with inconsistent effects on body weight, cardiometabolic parameters, symptoms, and functional performance.^{93,94} Importantly, studies evaluating lifestyle interventions in HFrEF have seldom enrolled individuals with obesity or incorporated weight loss-focused endpoints. Collectively, these limitations have resulted in modest and inconsistent effects of lifestyle interventions in real-world and HFrEF-specific settings, underscoring the need for adjunctive pharmacologic therapy.

BARIATRIC SURGERY. Bariatric surgery is an established intervention for sustained weight loss in individuals with a BMI ≥ 35 kg/m² or ≥ 30 kg/m² with obesity-related comorbidities.⁷ In HFrEF, interest in bariatric surgery has stemmed from its potential to improve eligibility for advanced therapies, particularly heart transplantation, for which a BMI > 35 kg/m² is regarded as a contraindication.⁹⁵ Although prospective data are lacking, retrospective studies suggest potential benefits in patients with HFrEF and obesity. An analysis of Medicare beneficiaries found that bariatric surgery was associated with reduced all-cause mortality,⁹⁶ whereas other studies have reported lower hospital readmission rates, improvements in LVEF, reductions in hospitalizations, enhanced survival, and improved transplant candidacy, including those supported with LVADs.⁹⁷⁻¹⁰³ These early but encouraging findings suggest that bariatric surgery may expand transplant eligibility and improve outcomes in patients with obesity and HFrEF. However, prospective trials are needed to confirm these benefits and define best practices for integrating bariatric surgery into advanced HFrEF care.

CLINICAL CONSIDERATIONS FOR OBESITY MANAGEMENT IN HFrEF

Accurate assessment of adiposity is essential in individuals with HFrEF, as BMI alone often fails to reflect cardiometabolic risk; therefore, routine evaluation should incorporate measures of central obesity such as waist circumference and waist-to-height ratio.⁶ In these patients, interpretation of weight change should account for volume status, because early reductions in body weight may reflect decongestion rather than loss of adipose tissue.¹⁰⁴ Documenting edema-free weight using clinical volume assessment, stability of diuretic dosing, and longitudinal congestion markers may help prevent

misattribution of benefit.¹⁰⁴ For patients with HFrEF undergoing structured weight-reduction strategies, clinical focus should be on body composition and functional status, with emphasis on reducing fat mass while preserving skeletal muscle and physical performance, as low muscle mass and sarcopenic obesity are associated with worse functional status and outcomes.^{105,106} This approach involves pairing caloric restriction with resistance-based physical activity and adequate protein intake and evaluating response using functional performance and patient-reported health status rather than weight alone.¹⁰⁵ Moreover, weight change may alter blood pressure, kidney function, and diuretic requirements, necessitating close monitoring and timely adjustment of contemporary HF regimens to avoid hypotension, kidney dysfunction, electrolyte abnormalities, or clinical destabilization. Given the current uncertainty regarding the safety and efficacy of pharmacologic weight loss therapies in HFrEF, we suggest that these agents should be used only after clear discussion among patients, caregivers, and clinicians, explicitly acknowledging uncertainty related to both safety and efficacy, with recommendations updated as dedicated clinical trial data evolve. Last, obesity management is also clinically relevant in advanced HF, as elevated BMI is associated with longer transplant wait times and worse posttransplant survival; accordingly, structured and closely monitored weight-management strategies are an important component of advanced HF care, with bariatric surgery considered in selected, clinically stable patients based on center expertise and available resources, while maximally tolerated GDMT is maintained.¹⁰⁷

WEIGHT LOSS PHARMACOTHERAPY: A GROWING FIELD WITH A CRITICAL GAP IN HFrEF

Weight loss pharmacotherapy has become a key treatment option for obesity, with an expanding array of FDA-approved drugs and investigational therapies targeting appetite, nutrient absorption, and metabolic processes.¹⁰⁸ However, despite these advancements, a substantial knowledge gap remains regarding their safety and efficacy in patients with HFrEF. Historically, concerns over cardiovascular safety have contributed to weight loss medications not being tested in HF populations.² Older agents, such as fenfluramine-phentermine and sibutramine, were withdrawn from the market because of their association with valvopathy and increased risks of myocardial infarction and stroke, and were never studied in HFrEF.⁷ Phentermine and other stimulant-

based therapies remain FDA approved for short-term use but are contraindicated in patients with cardiovascular disease because of their potential to elevate heart rate and blood pressure, increasing the risk of arrhythmias and HF decompensation.⁷ Although orlistat has demonstrated modest weight loss benefits with some functional improvements in small HFrEF studies, its gastrointestinal side effects limit widespread use.⁷ Trials evaluating combination therapies such as phentermine-topiramate and bupropion-naltrexone have excluded patients with advanced HF,¹⁰⁸ leaving their safety and efficacy in HFrEF largely unknown.

The emergence of GLP-1 RAs and dual GLP-1/glucose-dependent insulinotropic polypeptide (GIP) RAs has significantly expanded the landscape of weight loss pharmacotherapy, offering new potential for cardiovascular risk reduction in patients with HFpEF,^{70,71} which has informed recent recommendations indicating the use of GLP-1 RAs to reduce HF hospitalization in this population.¹⁰⁹ However, no equivalent trials have been conducted in HFrEF, and the existing evidence is limited to very small non-obese HFrEF trials and methodologically constrained secondary analyses.

The issue also extends to ongoing trials of pharmacologic weight loss therapies, in which patients with HFrEF remain underrepresented, as highlighted in [Table 2](#). The SURMOUNT-MMO (Study of Tirzepatide on the Reduction of Morbidity and Mortality in Adults with Obesity) trial aims to assess the impact of tirzepatide on cardiovascular outcomes in individuals with obesity and high cardiovascular risk; however, it does not specifically include patients with HFrEF. Although this trial is expected to provide valuable long-term insights into the effects of intentional weight loss on cardiovascular morbidity and mortality, it will not address the critical gap in evidence regarding the safety and efficacy of GLP-1 RA and GIP/GLP-1 co-agonists in individuals with HFrEF. In addition, there remains limited evidence on the role of weight loss medications in facilitating LVAD bridging or improving heart transplant eligibility among patients with severe obesity and advanced HFrEF.²

Beyond currently approved medications, several novel weight loss agents are in various stages of development, yet none are specifically being studied in HFrEF ([Table 2](#)). Cagrilintide/semaglutide, a combination of a GLP-1 RA and an amylin analogue, is currently being evaluated in a phase 3 trial assessing major adverse cardiovascular events, with HF outcomes included as secondary endpoints (REDEFINE 3: A Research Study to See the Effects of CagriSema in

TABLE 2 Ongoing and Pending Trials of Weight Loss Medications

Ongoing Trial	Weight Loss Medication	Mechanism of Action	Key Inclusion Criteria	Key Exclusion Criteria	HFrEF Inclusion/Exclusion Status ^a	Outcomes Assessed
SURMOUNT-MMO NCT05556512	Tirzepatide (LY3298176)	GLP-1/GIP dual RA	Age ≥40 years with BMI ≥27 kg/m ² and either established CVD (eg, CAD) or multiple CV risk factors	Type 1 or type 2 DM, recent major CV events (eg, acute decompensated HF within 90 days before screening), or significant GI, hepatic, pancreatic, or neoplastic conditions	HFrEF not specified; recent ADHF excluded (≤90 days)	Time to first occurrence of any component event of the composite (all-cause death, nonfatal MI, nonfatal stroke, coronary revascularization, or HF events) Changes in body weight, SBP, and DBP
TRIUMPH-3 NCT05882045	Retatrutide (LY3437943)	GLP-1/GIP/GCGR triple RA	Age ≥18 years with BMI ≥35 kg/m ² and established CVD; must have failed at least 1 prior dietary weight loss attempt	Recent CV events (eg, hospitalization for congestive HF within 90 days before screening); prior/ planned bariatric surgery, recent weight change >5 kg or use of weight loss drugs within 90 days, type 1 DM, history of pancreatitis or MEN-2 syndrome	HFrEF not specified, recent HF hospitalization excluded (≤90 days)	Changes in body weight, BMI, WC, and metabolic parameters
CAMEO-SEMA NCT05371496	Semaglutide	GLP-1 RA	Age ≥18 years with BMI ≥30 kg/m ² , NYHA functional class II-IV, and hospitalization for decompensated HF requiring IV loop diuretics within the past 12 months	MI, stroke, HF hospitalization, unstable angina, or TIA within 30 days before screening, SBP >160 mm Hg at screening, or planned coronary, carotid, or peripheral artery revascularization	HFrEF excluded as only people with LVEF ≥50% were eligible	Change in PCWP during exercise Changes in the parameters of cardiac remodeling and function Changes in body fat mass and visceral fat content Change in QoL assessed by the KCCQ
REDEFINE 3 NCT05669755	CagriSema (cagrilintide and semaglutide)	Amylin analogue and GLP-1 RA	Age ≥55 years with BMI ≥25 kg/m ² and established CVD, including a history of MI, stroke, or PAD	Recent MI, stroke, or hospitalization for unstable angina (<60 days before screening), severe kidney disease (eGFR <15 mL/min/1.73 m ²), or recent GLP-1 RA use (within 90 days of screening)	HFrEF not specified; NYHA functional class IV excluded	Time to first occurrence of MACE (CV death, nonfatal MI, and stroke) Changes in body weight, WC, SBP, DBP, and metabolic parameters
ACHIEVE-4 NCT05803421	Orforglipron (LY3502970)	Oral GLP-1RA	Adults ≥18 years with T2DM, HbA1c 7.0%-10.5%, BMI ≥25 kg/m ² , stable weight, and elevated CV risk defined by CHD, PAD, cerebrovascular disease, CKD, or CHF (NYHA functional class II-III)	Type 1 DM, use of weight loss drugs or supplements within 90 days of screening, or use of diabetes medications other than metformin, SGLT2i, or sulfonylureas	CVD allowed; HFrEF not specified	Time to first occurrence of MACE (MI, stroke, or CV death) Changes in body weight and metabolic parameters
SYNCHRONIZ - CVOT NCT06077864	Survodutide (BI 456906)	GLP-1/GCGR dual RA	BMI ≥27 kg/m ² with established CVD and/or ≥2 weight-related risk factors; or BMI ≥30 kg/m ² with CVD, CKD, or ≥2 cardiometabolic risk factors	Prior use of GLP-1 RAs within 3 months, type 1 diabetes, or significant gastric emptying disorders	CVD allowed; HFrEF not specified	Time to first adjudicated composite event (CV death, nonfatal MI, nonfatal stroke, ischemia-related coronary revascularization, or HF event) Changes in body weight, WC, SBP, and DBP Achievement of weight reduction ≥5%, ≥10%, ≥15% Change in KCCQ-TSS

Continued on the next page

TABLE 2 Continued

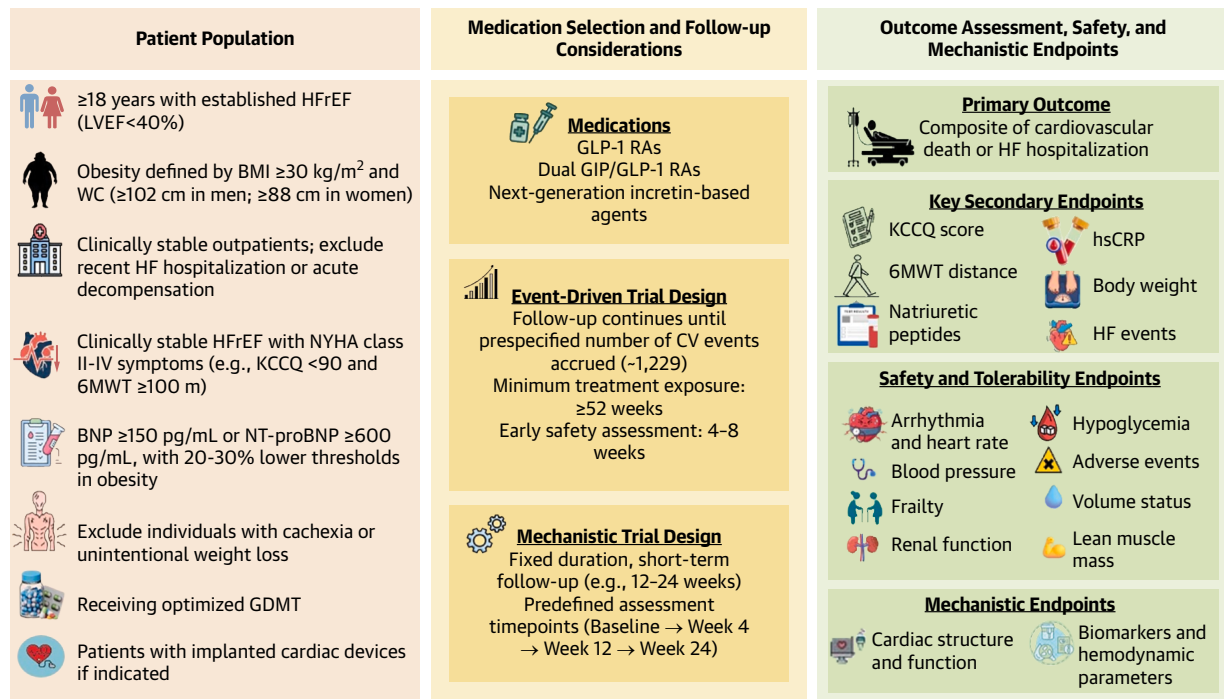
Ongoing Trial	Weight Loss Medication	Mechanism of Action	Key Inclusion Criteria	Key Exclusion Criteria	HFrEF Inclusion/Exclusion Status ^a	Outcomes Assessed
NCT05997576	TG103	GLP-1 RA	Age 18-75 years with BMI ≥ 28 kg/m ² , or 24-28 kg/m ² with ≥ 1 obesity-related complication; no weight loss or $< 5\%$ weight loss after lifestyle intervention in past 12 weeks	Diabetes of any type, secondary obesity, recent GLP-1 RA use within 12 weeks, bariatric surgery and severe GI disease	HFrEF not specified, but NYHA functional class I-III HF excluded	Changes in body weight, WC, and metabolic parameters Proportion of participants with weight loss of $\geq 5\%$ and $\geq 10\%$
GLORY-2 NCT06164873	Mazdutide (IBI362)	GLP-1/GCGR dual RA	BMI ≥ 30 kg/m ² with ≥ 1 unsuccessful dietary attempt. T2DM eligible if diagnosed ≥ 3 months prior, HbA1c 7.0%-10.0%, and on lifestyle or stable oral therapy	Recent weight change $> 5\%$ within 3 months or HbA1c $\geq 6.5\%$ at screening	HFrEF not specified	Changes in body weight, WC, SBP, and metabolic parameters Proportion of participants achieving body weight reduction (5%-20%)
NCT05934110	Orlistat/ acarbose (EMP16)	Orlistat: gastric/ pancreatic lipase inhibitor; acarbose: intestinal α -glucosidase inhibitor	Age ≥ 18 years with BMI ≥ 30 kg/m ² or ≥ 27 kg/m ² with ≥ 1 risk factor (eg, HTN, dyslipidemia, or T2DM treated with lifestyle only)	Weight change $\geq 5\%$, T2DM requiring medication, untreated high blood pressure, and GI conditions	HFrEF not specified	Changes in body weight, WC, BMI, body fat, and metabolic parameters Proportion of participants achieving $\geq 5\%$ decrease in body weight
DREAM NCT05788601	Dapaglutide	GLP-1/GLP-2 RA	Age 18-75 years with BMI ≥ 30 kg/m ² and history of at least 1 weight loss attempt	Weight change $\geq 5\%$ (past 90 days), DM, or HbA1c ≥ 48 mmol/mol, recent CV events (< 60 days), eGFR < 60 mL/min/1.73 m ² , recent use of obesity-related medications	HFrEF not specified but NYHA functional class IV HF excluded	Change in body weight Reduction in body weight (5%-15%)
MOMENTUM NCT05295875	Pemvidutide (ALT-801)	GLP-1/GCGR dual RA	Age 18-75 years, BMI ≥ 30 kg/m ² or ≥ 27 kg/m ² with ≥ 1 obesity-related comorbidity, and at least 1 unsuccessful weight loss attempt	Weight change $> 5\%$ in the past 3 months or $> 10\%$ in the past 6 months, any type of diabetes, genetic or endocrine obesity, or history of bariatric surgery or any other surgical treatment for obesity	HFrEF not specified	Changes in body weight and metabolic parameters Proportion of participants achieving $\geq 5\%$, $\geq 10\%$, $\geq 15\%$ weight loss

^aHFrEF inclusion and exclusion status reflects our interpretation based on a review of each trial's eligibility criteria.

ACHIEVE-4 = A Study of Daily Oral Orforglipron [LY3502970] Compared With Insulin Glargine in Participants With Type 2 Diabetes and Obesity or Overweight at Increased Cardiovascular Risk; ADHF = acute decompensated heart failure; CAD = coronary artery disease; CAMEO-SEMA = Cardiac and Metabolic Effects of Semaglutide in Heart Failure With Preserved Ejection Fraction; CHD = coronary heart disease; CKD = chronic kidney disease; CV = cardiovascular; CVD = cardiovascular disease; DBP = diastolic blood pressure; DM = diabetes mellitus; DREAM = Dapaglutide for the Treatment of Obesity; eGFR = estimated glomerular filtration rate; GCGR = glucagon receptor; GI = gastrointestinal; GIP = glucose-dependent insulinotropic polypeptide; GLORY-2 = A Study of IBI362 9 mg in Chinese Adults With Obesity; GLP-1 = glucagon-like peptide-1; GLP-1R = glucagon-like peptide-1 receptor; GLP-2 = glucagon-like peptide-2; HbA1c = hemoglobin A1c; HR = heart rate; HTN = hypertension; IV = intravenous; KCCQ-TSS = Kansas City Cardiomyopathy Questionnaire Total Symptom Score; MACE = major adverse cardiovascular events; MEN-2 = Multiple Endocrine Neoplasia type 2; MI = myocardial infarction; MOMENTUM = Efficacy and Safety of ALT-801 in the Treatment of Obesity; [NCT05997576](#) = A Study of TG103 Injection in Non-diabetic Overweight or Obesity; [NCT05934110](#) = Study Exploring the Supportive Effect of Acarbose in Weight Management; PAD = peripheral artery disease; PCWP = pulmonary capillary wedge pressure; RA = receptor agonist; REDEFINE 3 = A Research Study to See the Effects of CagriSema in People Living With Diseases in the Heart and Blood Vessels; SBP = systolic blood pressure; SGLT2i = sodium-glucose cotransporter-2 inhibitor; SURMOUNT-MMO = Study of Tirzepatide on the Reduction of Morbidity and Mortality in Adults with Obesity; SYNCHRONIZE™ - CVOT = A Study to Test the Effect of Survodutide [BI 456906] on Cardiovascular Safety in People With Overweight or Obesity ([NCT06077864](#)); T2DM = type 2 diabetes mellitus; TC = total cholesterol; TG = triglycerides; TIA = transient ischemic attack; TRIUMPH-3 = A Study of Retatrutide (LY3437943) in Participants With Obesity and Cardiovascular Disease; WC = waist circumference; other abbreviations as in [Table 1](#).

People Living With Diseases in the Heart and Blood Vessels; [NCT05669755](#)). Orforglipron, an oral small-molecule GLP-1 RA, is tested against insulin glargine in individuals with obesity, elevated cardiovascular risk, and T2DM (A Study of Daily Oral Orforglipron [LY3502970] Compared With Insulin Glargine in Participants With Type 2 Diabetes and Obesity or Overweight at Increased Cardiovascular Risk [ACHIEVE-4]; [NCT05803421](#)). Similarly, survodutide, a dual GLP-1 and glucagon receptor agonist, is

under phase 3 investigation in patients with obesity, established cardiovascular or kidney disease, and weight-related complications (A Study to Test the Effect of Survodutide [BI 456906] on Cardiovascular Safety in People With Overweight or Obesity [SYNCHRONIZE™ - CVOT]; [NCT06077864](#)). Although these studies include individuals with cardiometabolic risk, none are designed to evaluate safety or efficacy specifically in HFrEF. As the therapeutic landscape continues to expand, newer

FIGURE 2 Future Directions for Incretin-Based Weight Loss Therapies in HFrEF

This figure illustrates key considerations for future trials of incretin-based weight loss therapies in HFrEF, including patient population selection, medication selection, and follow-up duration. It also highlights priority outcome measures and safety endpoints relevant to this population. 6MWT = 6-minute walk test; BMI = body mass index; BNP = B-type natriuretic peptide; CV = cardiovascular; Dual GIP/GLP-1 RAs = dual glucose-dependent insulinotropic polypeptide and glucagon-like peptide-1 receptor agonists; GDMT = guideline-directed medical therapy; GLP-1 RAs = glucagon-like peptide-1 receptor agonists; hsCRP = high-sensitivity C-reactive protein; HF = heart failure; KCCQ = Kansas City Cardiomyopathy Questionnaire; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; WC = waist circumference.

pharmacologic classes including dual and triple incretin agonists as well as emerging agents such as amylin and PYY analogues warrant rigorous investigation to define their cardiovascular safety, therapeutic potential, and role in improving outcomes for patients with obesity and HFrEF.

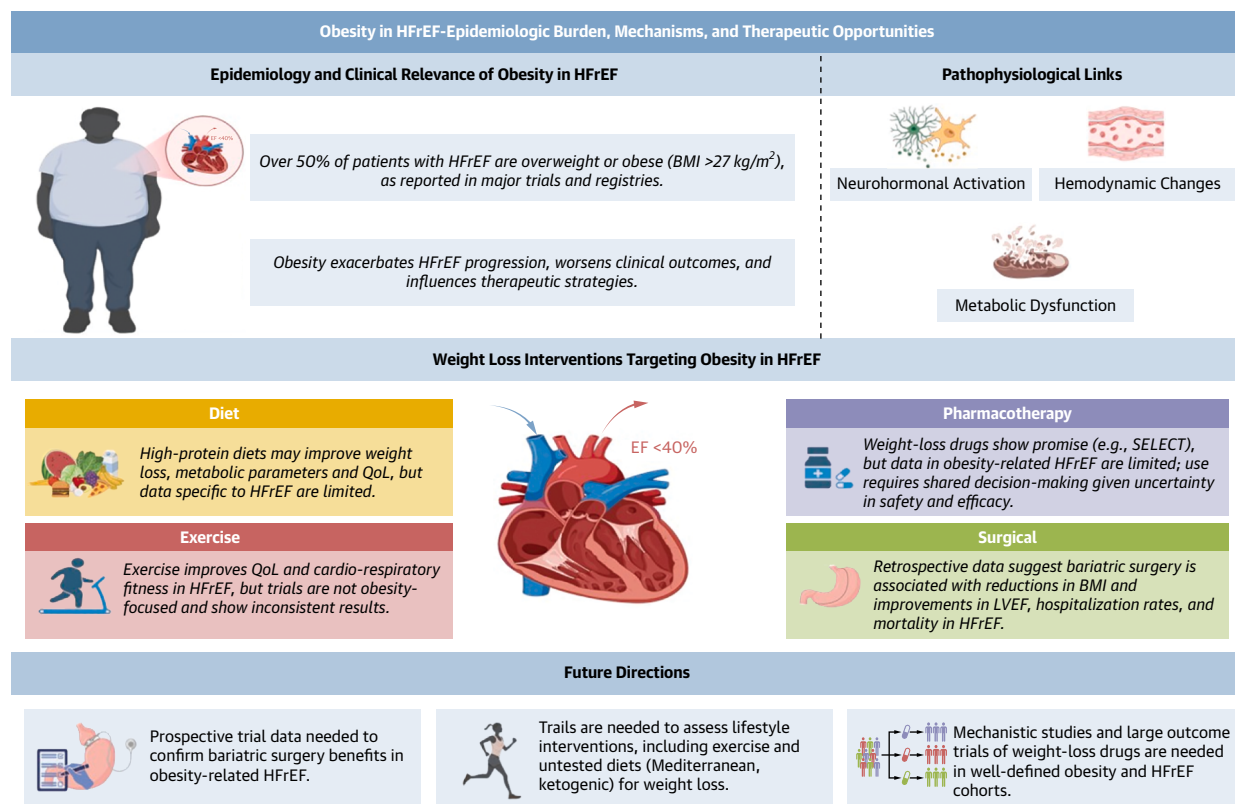
FUTURE DIRECTIONS FOR INCRETIN-BASED WEIGHT LOSS THERAPIES IN HFrEF

Given that obesity appears to modify the trajectory of HFrEF through multiple interacting pathophysiological pathways, incretin-based therapies such as GLP-1 RAs represent a biologically plausible strategy to target these mechanisms, providing a strong rationale for evaluation in future HFrEF trials. Future research should prioritize 2 key trial paradigms most likely to advance the field: event-driven outcome trials and mechanistic trials. Event-driven

outcome trials would help clarify whether incretin-based therapies improve clinically meaningful outcomes in HFrEF, including HF hospitalization, cardiovascular mortality, and patient-reported outcomes. Mechanistic trials would be useful in elucidating how these therapies influence cardiac structure and function, biomarker profiles, and hemodynamic parameters in patients with HFrEF. Together, these trial approaches, as outlined in **Figure 2**, can help address critical knowledge gaps and provide a framework for identifying patient subgroups most likely to benefit from incretin-based therapies in HFrEF.

CONCLUSIONS

Although obesity is highly prevalent among patients with HFrEF, its epidemiologic and pathophysiological relevance remains less well defined than in

CENTRAL ILLUSTRATION Overview of Obesity in Heart Failure With Reduced Ejection Fraction

Khan MS, et al. JACC. 2026;■(■):■-■.

This figure shows the high prevalence and clinical impact of obesity in HFrEF, as well as the associated pathophysiological pathways and current weight loss interventions, including dietary, exercise-based, pharmacological, and surgical approaches. It also highlights unmet needs and future directions for obesity-focused research in HFrEF. BMI = body mass index; EF = ejection fraction; HFrEF = heart failure with reduced ejection fraction; LVEF = left ventricular ejection fraction; QoL = quality of life; SELECT = Semaglutide Effects on Cardiovascular Outcomes in People With Overweight or Obesity.

HFpEF. Rather than serving primarily as a causal factor, obesity in HFrEF appears to shape disease course by amplifying existing pathophysiologic stress and influencing therapeutic options. Obesity-related mechanisms established in HFpEF may be relevant to HFrEF progression, albeit with differing relative contributions, providing a rationale to explore weight loss strategies, including GLP-1 RAs, in this population. However, evidence specific to HFrEF remains limited and is largely derived from non-obesity-focused trials, and patients with HFrEF have been largely excluded from contemporary anti-obesity medication trials. Dedicated, adequately powered studies are therefore needed to define the safety and clinical effects of anti-obesity medications and other weight loss interventions in HFrEF (**Central Illustration**).

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Khan has received fees from Bayer and Novartis. Dr Petrie has received research funding from Boehringer Ingelheim, Roche, SQ Innovations, AstraZeneca, Novartis, Novo Nordisk, Medtronic, Boston Scientific, and Pharmacosmos; and has consulted or participated on trial committees for Akero, Applied Therapeutics, Amgen, Ana-Cardio, Biosensors, Boehringer Ingelheim, Novartis, AstraZeneca, Novo Nordisk, AbbVie, Bayer, Horizon Therapeutics, Takeda, Cardiorientis, Pharmacosmos, Siemens, Eli Lilly, Vifor, New Amsterdam, Moderna, Teikoku, LIB Therapeutics, and 3R Lifesciences. Dr Zieroth has received research grant support, served on advisory boards for, or has had speaker engagements with Abbott, Akcea, AstraZeneca, Amgen, Alnylam, Bayer, Bristol Myers Squibb, Boehringer Ingelheim, Eli Lilly, GlaxoSmithKline, Janssen, Merck, Novartis, Novo Nordisk, Otsuka, Pfizer, Roche, Servier, and Vifor Pharma; and serves on a clinical trial committee or as a national lead for studies sponsored by AstraZeneca, Bayer, Boehringer Ingelheim, Merck, Novartis, and Pfizer. Dr Anker has received grants and personal fees from CSL Vifor and Abbott Vascular; has received personal fees for consultancies, trial committee work, and/or lectures from Actimed, AstraZeneca,

Bayer, Bioventrix, Boehringer Ingelheim, Brahms, Cardiac Dimensions, Cardior, Cordio, CVRx, Cytokinetics, Edwards, Farraday Pharmaceuticals, GSK, HeartKinetics, Impulse Dynamics, Medtronic, Novartis, Novo Nordisk, Occlutech, Pfizer, Regeneron, Relaxera, Repairon, Scirent, Sensible Medical, Servier, Vectorious, and V-Wave; and was named co-inventor of 2 patent applications regarding MR-proANP (DE 102007010834 and DE 102007022367) but does not benefit personally from the related issued patents. Dr Butler has received consulting fees from Abbott, American Regent, Amgen, Applied Therapeutic, AskBio, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, Boston Scientific, Bristol Myers Squibb, Cardiac Dimension, Cardiocell, Cardior, Cardiore, CSL Bearing, CVRx, Cytokinetics, Daxor, Edwards, Element Science, Faraday, Foundry, G3P, Innolife, Impulse Dynamics, Imbria, Inventiva, Ionis, Lexicon, Lilly, LivaNova, Janssen, Medtronic, Merck, Occlutech, Owkin, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Pfizer, Prolaio, Regeneron, Renibus, Roche, Salamandra, Sanofi, SC Pharma,

Secretome, Sequana, SQ Innovation, Tenex, Tricog, Ultralytics, CSL Vifor, and Zoll; and receives honoraria from Novartis, Boehringer Ingelheim-Lilly, AstraZeneca, Impulse Dynamics, and CSL Vifor. Dr Javid has reported that he has no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Muhammad Shahzeb Khan, The Heart Hospital Plano, 1100 Allied Drive, Plano, Texas 75093, USA. E-mail: Shahzeb.khan@bswhealth.org. OR Dr Javed Butler, Baylor Scott and White Research Institute, 3434 Live Oak Street, Suite 501, Dallas, Texas 75204, USA. E-mail: javed.butler@bswhealth.org.

REFERENCES

- Elagizi A, Carbone S, Lavie CJ, Mehra MR, Ventura HO. Implications of obesity across the heart failure continuum. *Prog Cardiovasc Dis*. 2020;63(5):561-569. <https://doi.org/10.1016/j.pcad.2020.09.005>
- Harrington J, Gale SE, Vest AR. Anti-obesity medications in patients with heart failure: current evidence and practical guidance. *Circ Heart Fail*. 2024;17(9):e011518. <https://doi.org/10.1161/CIRCHEARTFAILURE.124.011518>
- Butler J, Nair A, Khan MS. Body mass index distribution among patients with heart failure and reduced ejection fraction: implication for future trials. *J Card Fail*. 2024;30(7):958-960. <https://doi.org/10.1016/j.cardfail.2024.03.009>
- DeVore AD, Mi X, Thomas L, et al. Characteristics and treatments of patients enrolled in the CHAMP-HF registry compared with patients enrolled in the PARADIGM-HF trial. *J Am Heart Assoc*. 2018;7(12):e009237. <https://doi.org/10.1161/JAHA.118.009237>
- Chioncel O, Lainscak M, Seferovic PM, et al. Epidemiology and one-year outcomes in patients with chronic heart failure and preserved, mid-range and reduced ejection fraction: an analysis of the ESC Heart Failure Long-Term Registry. *Eur J Heart Fail*. 2017;19(12):1574-1585. <https://doi.org/10.1002/ehf.813>
- Kittleson MM, Benjamin EJ, Blumer V, et al. 2025 ACC scientific statement on the management of obesity in adults with heart failure: a report of the American College of Cardiology. *J Am Coll Cardiol*. 2025;86(20):1953-1975. <https://doi.org/10.1016/j.jacc.2025.05.008>
- Vest AR, Schauer PR, Rodgers JE, et al. Obesity and weight loss strategies for patients with heart failure. *JACC Heart Fail*. 2024;12(9):1509-1527. <https://doi.org/10.1016/j.jchf.2024.06.006>
- Packer M. Do obesity and visceral adiposity promote heart failure with reduced ejection fraction? *Eur Heart J*. 2026;47(1):12-21. <https://doi.org/10.1093/eurheartj/ehaf645>
- Sagar UN, Ahmed MM, Adams S, Whellan DJ. Does body mass index really matter in the management of heart failure? A review of the literature. *Cardiol Rev*. 2008;16(3):124-128. <https://doi.org/10.1097/CRD.0b013e31815d29e9>
- Savji N, Meijers WC, Bartz TM, et al. The association of obesity and cardiometabolic traits with incident HFpEF and HFrEF. *JACC Heart Fail*. 2018;6(8):701-709. <https://doi.org/10.1016/j.jchf.2018.05.018>
- Pandey A, LaMonte M, Klein L, et al. Relationship between physical activity, body mass index, and risk of heart failure. *J Am Coll Cardiol*. 2017;69(9):1129-1142. <https://doi.org/10.1016/j.jacc.2016.11.081>
- Eaton CB, Pettinger M, Rossouw J, et al. Risk factors for incident hospitalized heart failure with preserved versus reduced ejection fraction in a multiracial cohort of postmenopausal women. *Circ Heart Fail*. 2016;9(10):e002883. <https://doi.org/10.1161/CIRCHEARTFAILURE.115.002883>
- Rao VN, Zhao D, Allison MA, et al. Adiposity and incident heart failure and its subtypes: MESA (Multi-Ethnic Study of Atherosclerosis). *JACC Heart Fail*. 2018;6(12):999-1007. <https://doi.org/10.1016/j.jchf.2018.07.009>
- Robertson J, Lindgren M, Schaufelberger M, et al. Body mass index in young women and risk of cardiomyopathy: a long-term follow-up study in Sweden. *Circulation*. 2020;141(7):520-529. <https://doi.org/10.1161/CIRCULATIONAHA.119.044056>
- Robertson J, Schaufelberger M, Lindgren M, et al. Higher body mass index in adolescence predicts cardiomyopathy risk in midlife. *Circulation*. 2019;140(2):117-125. <https://doi.org/10.1161/CIRCULATIONAHA.118.039132>
- Alraies MC, Eckman P. Adult heart transplant: indications and outcomes. *J Thorac Dis*. 2014;6:1120-1128. <https://doi.org/10.3978/j.issn.2072-1439.2014.06.44>
- Butler J, Arshad MS, Khan MS. Role of anti-obesity drugs in heart failure regardless of ejection fraction. *Eur J Heart Fail*. 2024;26(8):1703-1706. <https://doi.org/10.1002/ehf.3224>
- Packer M. Leptin-aldosterone-nephrilysin axis: identification of its distinctive role in the pathogenesis of the three phenotypes of heart failure in people with obesity. *Circulation*. 2018;137(15):1614-1631. <https://doi.org/10.1161/CIRCULATIONAHA.117.032474>
- Powell-Wiley TM, Poirier P, Burke LE, et al. Obesity and cardiovascular disease: a scientific statement from the American Heart Association. *Circulation*. 2021;143(21):e984-e1010. <https://doi.org/10.1161/CIR.0000000000000973>
- Ho JE, Lyass A, Lee DS, et al. Predictors of new-onset heart failure: differences in preserved versus reduced ejection fraction. *Circ Heart Fail*. 2013;6(2):279-286. <https://doi.org/10.1161/CIRCHEARTFAILURE.112.972828>
- Kodama S, Fujihara K, Horikawa C, et al. Diabetes mellitus and risk of new-onset and recurrent heart failure: a systematic review and meta-analysis. *ESC Heart Fail*. 2020;7(5):2146-2174. <https://doi.org/10.1002/ehf2.12782>
- Hartup J, Mann DL. Neurohormonal activation in heart failure with reduced ejection fraction. *Nat Rev Cardiol*. 2017;14(1):30-38. <https://doi.org/10.1038/nrcardio.2016.163>
- Matsumoto S, Butt JH, Docherty KF, et al. Obesity, aldosterone, and natriuretic peptide in patients with heart failure and reduced ejection fraction. *Cardiovasc Res*. Published online November 18, 2025. <https://doi.org/10.1093/cvr/cvaf235>
- Theodorakis N, Nikolaou M. Leptin and heart failure: the chicken or the egg? *Heart Fail Rev*. 2025;30(4):749-757. <https://doi.org/10.1007/s10741-025-10501-6>
- Hansen BL, Deis T, Larsson JE, et al. Influence of obesity on invasive hemodynamics and prognosis in patients with heart failure. *JACC Heart*

- Fail. 2025;13(5):725-736. <https://doi.org/10.1016/j.jchf.2024.12.009>
26. Reddy YNV, Melenovsky V, Redfield MM, Nishimura RA, Borlaug BA. High-output heart failure: a 15-year experience. *J Am Coll Cardiol*. 2016;68(5):473-482. <https://doi.org/10.1016/j.jacc.2016.05.043>
 27. Kasper EK, Hruban RH, Baughman KL. Cardiomyopathy of obesity: a clinicopathologic evaluation of 43 obese patients with heart failure. *Am J Cardiol*. 1992;70(9):921-924. [https://doi.org/10.1016/0002-9149\(92\)90739-L](https://doi.org/10.1016/0002-9149(92)90739-L)
 28. Hofmann U, Heuer S, Meder K, et al. The proinflammatory cytokines TNF-alpha and IL-1 beta impair economy of contraction in human myocardium. *Cytokine*. 2007;39(3):157-162. <https://doi.org/10.1016/j.cyto.2007.07.185>
 29. Pugliese NR, Paneni F, Mazzola M, et al. Impact of epicardial adipose tissue on cardiovascular haemodynamics, metabolic profile, and prognosis in heart failure. *Eur J Heart Fail*. 2021;23(11):1858-1871. <https://doi.org/10.1002/ehf.2337>
 30. Requena-Ibáñez JA, Santos-Gallego CG, Rodriguez-Cordero A, et al. Mechanistic insights of empagliflozin in nondiabetic patients with HFREF: from the EMPA-TROPISM study. *JACC Heart Fail*. 2021;9(8):578-589. <https://doi.org/10.1016/j.jchf.2021.04.014>
 31. Takano H, Obata JE, Kodama Y, et al. Adiponectin is released from the heart in patients with heart failure. *Int J Cardiol*. 2009;132(2):221-226. <https://doi.org/10.1016/j.ijcard.2007.11.040>
 32. Gutiérrez-Cuevas J, Sandoval-Rodríguez A, Meza-Ríos A, et al. Molecular mechanisms of obesity-linked cardiac dysfunction: an up-date on current knowledge. *Cells*. 2021;10(3):629. <https://doi.org/10.3390/cells10030629>
 33. Ren J, Wu NN, Wang S, Sowers JR, Zhang Y. Obesity cardiomyopathy: evidence, mechanisms, and therapeutic implications. *Physiol Rev*. 2021;101(4):1745-1807. <https://doi.org/10.1152/physrev.00030.2020>
 34. Rayner JJ, Banerjee R, Holloway CJ, et al. The relative contribution of metabolic and structural abnormalities to diastolic dysfunction in obesity. *Int J Obes (Lond)*. 2018;42(3):441-447. <https://doi.org/10.1038/s41301-017-239>
 35. Neglia D, De Caterina A, Marraccini P, et al. Impaired myocardial metabolic reserve and substrate selection flexibility during stress in patients with idiopathic dilated cardiomyopathy. *Am J Physiol Heart Circ Physiol*. 2007;293(6):H3270-H3278. <https://doi.org/10.1152/ajpheart.00887.2007>
 36. Rayner JJ, Peterzan MA, Watson WD, et al. Myocardial energetics in obesity: enhanced ATP delivery through creatine kinase with blunted stress response. *Circulation*. 2020;141(14):1152-1163. <https://doi.org/10.1161/CIRCULATIONAHA.119.042770>
 37. Castillo EC, Morales JA, Chapoy-Villanueva H, et al. Mitochondrial hyperacetylation in the failing hearts of obese patients mediated partly by a reduction in SIRT3: the involvement of the mitochondrial permeability transition pore. *Cell Physiol Biochem*. 2019;53(3):465-479. <https://doi.org/10.33594/000000151>
 38. Wilding JPH, Batterham RL, Calanna S, et al. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med*. 2021;384(11):989-1002. <https://doi.org/10.1056/NEJMoa2032183>
 39. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387(3):205-216. <https://doi.org/10.1056/NEJMoa2206038>
 40. Pi-Sunyer X, Astrup A, Fujioka K, et al. A randomized, controlled trial of 3.0 mg of liraglutide in weight management. *N Engl J Med*. 2015;373(1):11-22. <https://doi.org/10.1056/NEJMoa1411892>
 41. Puglisi S, Rossini A, Poli R, et al. Effects of SGLT2 inhibitors and GLP-1 receptor agonists on renin-angiotensin-aldosterone system. *Front Endocrinol (Lausanne)*. 2021;12:738848. <https://doi.org/10.3389/fendo.2021.738848>
 42. Moiz A, Zolotarova T, Filion KB, Eisenberg MJ. GLP-1 receptor agonists and blood pressure: a state-of-the-art review of mechanisms, evidence, and clinical implications. *Am J Hypertens*. Published online October 23, 2025. <https://doi.org/10.1093/ajh/hpaf205>
 43. Chen J, Xiao R, Gao L, et al. Real-world challenges for guideline-directed medical therapy intolerance in heart failure: a single-center prospective cohort study. *Int J Cardiol*. 2025;434:133367. <https://doi.org/10.1016/j.ijcard.2025.133367>
 44. Borlaug BA, Holve C, Jung MH, et al. Prevalence and associations of systemic inflammation in heart failure across the spectrum of ejection fraction. *JACC Heart Fail*. 2026;14(2):102712. <https://doi.org/10.1016/j.jchf.2025.102712>
 45. Edwards M, Thomas M, Farr M, et al. Glucagon-like peptide-1 agonist use for obesity treatment in patients with left ventricular assist devices. *JHLT Open*. 2024;5:100114. <https://doi.org/10.1016/j.jhlto.2024.100114>
 46. daSilva-deAbreu A, Rodgers JE, Seltz J, et al. Obesity, challenges, and weight-loss strategies for patients with ventricular assist devices. *JACC Heart Fail*. 2024;12(10):1661-1676. <https://doi.org/10.1016/j.jchf.2024.04.006>
 47. Gadela NV, Umashanker D, Scatola A, Jaiswal A. Clinical outcomes, trends in weight, and weight loss strategies in patients with obesity after durable ventricular assist device implantation. *Curr Heart Fail Rep*. 2021;18(2):52-63. <https://doi.org/10.1007/s11897-020-00500-z>
 48. Kim SE, Lee CJ. The Paradox in defining obesity in patients with heart failure. *Int J Heart Fail*. 2022;4(2):91-94. <https://doi.org/10.36628/ijhf.2022.0010>
 49. Catalfamo VJ, Tutor AW, Koos A, Vyas A, Lavie CJ, Carbone S. Obesity, metabolic syndrome, and obesity paradox in heart failure: a critical evaluation. *Curr Heart Fail Rep*. 2024;12(1):1. <https://doi.org/10.1007/s11897-024-00690-w>
 50. Alrob OA, Sankaralingam S, Alazzam S, Nusairat B, Qattoum M, Nusair MB. Obesity paradox among heart failure with reduced ejection fraction patients: a retrospective cohort study. *Medicina (Kaunas)*. 2022;59(1):60. <https://doi.org/10.3390/medicina59010060>
 51. Heymsfield SB, Peterson CM, Thomas DM, Heo M, Schuna JM Jr. Why are there race/ethnic differences in adult body mass index-adiposity relationships? A quantitative critical review. *Obes Rev*. 2016;17(3):262-275. <https://doi.org/10.1111/obr.12358>
 52. Okorodudu DO, Jumeau MF, Montori VM, et al. Diagnostic performance of body mass index to identify obesity as defined by body adiposity: a systematic review and meta-analysis. *Int J Obes (Lond)*. 2010;34(5):791-799. <https://doi.org/10.1038/sj.ijo.2010.5>
 53. Piché ME, Poirier P, Lemieux I, Després JP. Overview of epidemiology and contribution of obesity and body fat distribution to cardiovascular disease: an update. *Prog Cardiovasc Dis*. 2018;61(2):103-113. <https://doi.org/10.1016/j.pcad.2018.06.004>
 54. Rothman KJ. BMI-related errors in the measurement of obesity. *Int J Obes (Lond)*. 2008;32(Suppl 3):S56-S59. <https://doi.org/10.1038/sj.ijo.2008.87>
 55. Butt JH, Petrie MC, Jhund PS, et al. Anthropometric measures and adverse outcomes in heart failure with reduced ejection fraction: revisiting the obesity paradox. *Eur Heart J*. 2023;44(13):1136-1153. <https://doi.org/10.1093/eurheartj/ehad083>
 56. Butt JH, Thune JJ, Nielsen JC, et al. Anthropometric measures and long-term mortality in non-ischaemic heart failure with reduced ejection fraction: questioning the obesity paradox. *Eur J Heart Fail*. 2025;27(3):527-536. <https://doi.org/10.1002/ehf.3424>
 57. Clark AL, Fonarow GC, Horwich TB. Waist circumference, body mass index, and survival in systolic heart failure: the obesity paradox revisited. *J Card Fail*. 2011;17(5):374-380. <https://doi.org/10.1016/j.cardfail.2011.01.009>
 58. Anker SD, Khan MS, Butler J, et al. Weight change and clinical outcomes in heart failure with reduced ejection fraction: insights from EMPEROR-Reduced. *Eur J Heart Fail*. 2023;25(1):117-127. <https://doi.org/10.1002/ehf.2728>
 59. Harrington J, Felker GM, Lingvay I, Pagidipati NJ, Pandey A, McGuire DK. Managing obesity in heart failure: a chance to tip the scales? *JACC Heart Fail*. 2024;12(1):28-34. <https://doi.org/10.1016/j.jchf.2023.09.005>
 60. McDowell K, Petrie MC, Raihan NA, Logue J. Effects of intentional weight loss in patients with obesity and heart failure: a systematic review. *Obes Rev*. 2018;19(9):1189-1204. <https://doi.org/10.1111/obr.12707>
 61. Jorsal A, Kistorp C, Holmager P, et al. Effect of liraglutide, a glucagon-like peptide-1 analogue, on left ventricular function in stable chronic heart failure patients with and without diabetes (LIVE)-a multicentre, double-blind, randomised, placebo-controlled trial. *Eur J Heart Fail*. 2017;19(1):69-77. <https://doi.org/10.1002/ehf.657>
 62. Margulies KB, Hernandez AF, Redfield MM, et al. Effects of liraglutide on clinical stability among patients with advanced heart failure and

- reduced ejection fraction: a randomized clinical trial. *JAMA*. 2016;316(5):500-508. <https://doi.org/10.1001/jama.2016.10260>
63. Neves JS, Vasques-Nóvoa F, Borges-Canha M, et al. Risk of adverse events with liraglutide in heart failure with reduced ejection fraction: a post hoc analysis of the FIGHT trial. *Diabetes Obes Metab*. 2023;25(1):189-197. <https://doi.org/10.1111/dom.14862>
 64. Neves JS, Leite AR, Mentz RJ, et al. Cardiovascular outcomes with exenatide in type 2 diabetes according to ejection fraction: the EXSCEL trial. *Eur J Heart Fail*. 2025;27(3):540-551. <https://doi.org/10.1002/ehf.3478>
 65. Neves JS, Packer M, Ferreira JP. Increased risk of heart failure hospitalization with GLP-1 receptor agonists in patients with reduced ejection fraction: a meta-analysis of the EXSCEL and FIGHT trials. *J Card Fail*. 2023;29(7):1107-1109. <https://doi.org/10.1016/j.cardfail.2023.03.017>
 66. Lepore JJ, Olson E, Demopoulos L, et al. Effects of the novel long-acting GLP-1 agonist, albiglutide, on cardiac function, cardiac metabolism, and exercise capacity in patients with chronic heart failure and reduced ejection fraction. *JACC Heart Fail*. 2016;4(7):559-566. <https://doi.org/10.1016/j.jchf.2016.01.008>
 67. Zhang JY, Wang XY, Wang X. Effects of liraglutide on hemodynamic parameters in patients with heart failure. *Oncotarget*. 2017;8(37):62693-62702. <https://doi.org/10.18632/oncotarget.18570>
 68. Lincoff AM, Brown-Frandsen K, Colhoun HM, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *N Engl J Med*. 2023;389(24):2221-2232. <https://doi.org/10.1056/NEJMoa2307563>
 69. Deanfield J, Verma S, Scirica BM, et al. Semaglutide and cardiovascular outcomes in patients with obesity and prevalent heart failure: a pre-specified analysis of the SELECT trial. *Lancet*. 2024;404(10454):773-786. [https://doi.org/10.1016/S0140-6736\(24\)01498-3](https://doi.org/10.1016/S0140-6736(24)01498-3)
 70. Kosiborod MN, Abildstrøm SZ, Borlaug BA, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2023;389(12):1069-1084. <https://doi.org/10.1056/NEJMoa2306963>
 71. Kosiborod MN, Petrie MC, Borlaug BA, et al. Semaglutide in patients with obesity-related heart failure and type 2 diabetes. *N Engl J Med*. 2024;390(15):1394-1407. <https://doi.org/10.1056/NEJMoa2313917>
 72. Packer M, Zile MR, Kramer CM, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2025;392(5):427-437. <https://doi.org/10.1056/NEJMoa2410027>
 73. Müller-Werdan U, Stöckl G, Werdan K. Advances in the management of heart failure: the role of ivabradine. *Vasc Health Risk Manag*. 2016;12:453-470. <https://doi.org/10.2147/VHRM.S90383>
 74. Masarone D, Martucci ML, Errigo V, Pacileo G. The use of β -blockers in heart failure with reduced ejection fraction. *J Cardiovasc Dev Dis*. 2021;8(9):101. <https://doi.org/10.3390/jcdd8090101>
 75. Khan MS, Fonarow GC, McGuire DK, et al. Glucagon-like peptide 1 receptor agonists and heart failure: the need for further evidence generation and practice guidelines optimization. *Circulation*. 2020;142(12):1205-1218. <https://doi.org/10.1161/CIRCULATIONAHA.120.045888>
 76. Pyke C, Heller RS, Kirk RK, et al. GLP-1 receptor localization in monkey and human tissue: novel distribution revealed with extensively validated monoclonal antibody. *Endocrinology*. 2014;155(4):1280-1290. <https://doi.org/10.1210/en.2013-1934>
 77. Packer M. Is the way to someone's heart through their stomach? The cardiorenal paradox of incretin-based hypoglycemic drugs in heart failure. *Circ Heart Fail*. 2017;10(10):e004551. <https://doi.org/10.1161/CIRCHEARTFAILURE.117.004551>
 78. Packer M, Carver JR, Rodeheffer RJ, et al. Effect of oral milrinone on mortality in severe chronic heart failure. The PROMISE Study Research Group. *N Engl J Med*. 1991;325(21):1468-1475. <https://doi.org/10.1056/NEJM199111213252103>
 79. Beck-da-Silva L, Higginson L, Fraser M, Williams K, Haddad H. Effect of Orlistat in obese patients with heart failure: a pilot study. *Congest Heart Fail*. 2005;11(3):118-123. <https://doi.org/10.1111/j.1527-5299.2005.03827.x>
 80. Anker SD, Ji L, Kindel T, et al. iCARDIO Alliance Global Implementation Guidelines for the Management of Obesity 2025—Focus on prevention and treatment of cardiometabolic disease. *Global Cardiol*. 2025;3(3). <https://doi.org/10.4081/cardio.2025.86>
 81. Horwich TB, Broderick S, Chen L, et al. Relation among body mass index, exercise training, and outcomes in chronic systolic heart failure. *Am J Cardiol*. 2011;108(12):1754-1759. <https://doi.org/10.1016/j.amjcard.2011.07.051>
 82. Groennebaek T, Sijlacks P, Nielsen R, et al. Effect of blood flow restricted resistance exercise and remote ischemic conditioning on functional capacity and myocardial adaptations in patients with heart failure. *Circ Heart Fail*. 2019;12(12):e006427. <https://doi.org/10.1161/CIRCHEARTFAILURE.119.006427>
 83. Corvera-Tindel T, Doering LV, Woo MA, Khan S, Dracup K. Effects of a home walking exercise program on functional status and symptoms in heart failure. *Am Heart J*. 2004;147(2):339-346. <https://doi.org/10.1016/j.ahj.2003.09.007>
 84. Lan NSR, Lam K, Naylor LH, et al. The impact of distinct exercise training modalities on echocardiographic measurements in patients with heart failure with reduced ejection fraction. *J Am Soc Echocardiogr*. 2020;33(2):148-156. <https://doi.org/10.1016/j.echo.2019.09.012>
 85. Gademan MG, Swenne CA, Verwey HF, et al. Exercise training increases oxygen uptake efficiency slope in chronic heart failure. *Eur J Cardiovasc Prev Rehabil*. 2008;15(2):140-144. <https://doi.org/10.1097/HJR.0b013e3282ef19986>
 86. de Meirelles LR, Matsuura C, Resende Ade C, et al. Chronic exercise leads to antiaggregant, antioxidant and anti-inflammatory effects in heart failure patients. *Eur J Prev Cardiol*. 2014;21(10):1225-1232. <https://doi.org/10.1177/2047487313491662>
 87. Dalal HM, Taylor RS, Jolly K, et al. The effects and costs of home-based rehabilitation for heart failure with reduced ejection fraction: The REACH-HF multicentre randomized controlled trial. *Eur J Prev Cardiol*. 2019;26(3):262-272. <https://doi.org/10.1177/2047487318806358>
 88. Piotrowicz E, Pencina MJ, Opolski G, et al. Effects of a 9-week hybrid comprehensive tele-rehabilitation program on long-term outcomes in patients with heart failure: The Telerehabilitation in Heart Failure Patients (TELEREH-HF) randomized clinical trial. *JAMA Cardiol*. 2020;5(3):300-308. <https://doi.org/10.1001/jamacardio.2019.5006>
 89. Kerrigan DJ, Williams CT, Ehrman JK, et al. Cardiac rehabilitation improves functional capacity and patient-reported health status in patients with continuous-flow left ventricular assist devices: the Rehab-VAD randomized controlled trial. *JACC Heart Fail*. 2014;2(6):653-659. <https://doi.org/10.1016/j.jchf.2014.06.011>
 90. Feuerstein A, Schoenrath F, Belyavskiy E, et al. Supervised exercise training in patients with advanced heart failure and left ventricular assist device: a multicentre randomized controlled trial (Ex-VAD trial). *Eur J Heart Fail*. 2023;25(12):2252-2262. <https://doi.org/10.1002/ehf.3032>
 91. Evangelista LS, Doering LV, Lennie T, et al. Usefulness of a home-based exercise program for overweight and obese patients with advanced heart failure. *Am J Cardiol*. 2006;97(6):886-890. <https://doi.org/10.1016/j.amjcard.2005.10.025>
 92. Vetrovsky T, Siranec M, Frybova T, et al. Lifestyle walking intervention for patients with heart failure with reduced ejection fraction: the WATCHFUL trial. *Circulation*. 2024;149(3):177-188. <https://doi.org/10.1161/CIRCULATIONAHA.123.067395>
 93. Evangelista LS, Heber D, Li Z, Bowerman S, Hamilton MA, Fonarow GC. Reduced body weight and adiposity with a high-protein diet improves functional status, lipid profiles, glycemic control, and quality of life in patients with heart failure: a feasibility study. *J Cardiovasc Nurs*. 2009;24(3):207-215. <https://doi.org/10.1097/JCN.0b013e31819846b9>
 94. Pritchett AM, Deswal A, Aguilar D, et al. Lifestyle modification with diet and exercise in obese patients with heart failure - a pilot study. *J Obes Weight Loss Ther*. 2012;2(2):1-8. <https://doi.org/10.4172/2165-7904.1000118>
 95. Sher T, Noom M, Ganam S, Sujka J, Rinde-Hoffman D, DuCoin C. Temporary mechanical circulatory support during bariatric surgery: a novel bridge to durable left ventricular assist device and cardiac transplantation. *Am Surg*. 2025;91(8):1273-1275. <https://doi.org/10.1177/00031348241244637>
 96. Mentias A, Desai MY, Aminian A, et al. Trends and outcomes associated with bariatric surgery and pharmacotherapies with weight loss effects among patients with heart failure and obesity. *Circ Heart Fail*. 2024;17(2):e010453. <https://doi.org/10.1161/CIRCHEARTFAILURE.122.010453>
 97. Ramani GV, McCloskey C, Ramanathan RC, Mathier MA. Safety and efficacy of bariatric surgery in morbidly obese patients with severe

systolic heart failure. *Clin Cardiol*. 2008;31(11):516-520. <https://doi.org/10.1002/clc.20315>

98. Sarmiento-Cobos M, Fonnegra CB, Montorfano L, et al. Short-term rapid weight loss induced by bariatric surgery improves ventricular ejection fraction in patients with severe obesity and heart failure. *Surg Obes Relat Dis*. 2021;17(9):1616-1620. <https://doi.org/10.1016/j.soard.2021.05.004>

99. Vest AR, Patel P, Schauer PR, et al. Clinical and echocardiographic outcomes after bariatric surgery in obese patients with left ventricular systolic dysfunction. *Circ Heart Fail*. 2016;9(3):e002260. <https://doi.org/10.1161/CIRCHEARTFAILURE.115.002260>

100. Goubar T, Kim S, Cistulli D, et al. Sleeve gastrectomy as a bridge to cardiac recovery - A retrospective comparative cohort study. *JHLT Open*. 2025;8:100224. <https://doi.org/10.1016/j.jhlto.2025.100224>

101. Zenilman A, Pechman D, Moran-Atkin E, Choi J, Camacho D. Bariatric surgery in patients with left ventricular assist devices: a safe and effective method of weight loss as a gateway to heart transplantation. *Surg Obes Relat Dis*. 2019;15(10):1780-1784. <https://doi.org/10.1016/j.soard.2019.08.003>

102. Hawkins RB, Go K, Raymond SL, Ayzengart A, Friedman J. Laparoscopic sleeve gastrectomy in patients with heart failure and left ventricular assist devices as a bridge to transplant. *Surg Obes Relat Dis*. 2018;14(9):1269-1273. <https://doi.org/10.1016/j.soard.2018.04.005>

103. Jeng EI, Miller AH, Friedman J, et al. Ventricular assist device implantation and bariatric surgery: a route to transplantation in morbidly obese patients with end-stage heart failure. *ASAIO J*. 2021;67(2):163-168. <https://doi.org/10.1097/MAT.0000000000001212>

104. Pena DL, Ilieșiu AM, Aurelian J, et al. Assessment of decongestion status before discharge in acute decompensated heart failure: a review of clinical, biochemical, and imaging tools and their impact on management decisions. *Medicina (Kaunas)*. 2025;61(5):816. <https://doi.org/10.3390/medicina61050816>

105. Caturano A, Amaro A, Berra CC, Conte C. Sarcopenic obesity and weight loss-induced muscle mass loss. *Curr Opin Clin Nutr Metab Care*. 2025;28(4):339-350. <https://doi.org/10.1097/MCO.0000000000001131>

106. Billingsley HE, Del Buono MG, Canada JM, et al. Sarcopenic obesity is associated with

reduced cardiorespiratory fitness compared with nonsarcopenic obesity in patients with heart failure with reduced ejection fraction. *Circ Heart Fail*. 2022;15(10):e009518. <https://doi.org/10.1161/CIRCHEARTFAILURE.122.009518>

107. Peled Y, Ducharme A, Kittleson M, et al. International Society for Heart and Lung Transplantation guidelines for the evaluation and care of cardiac transplant candidates-2024. *J Heart Lung Transplant*. 2024;43(10):1529-1628.e54. <https://doi.org/10.1016/j.healun.2024.05.010>

108. Melson E, Ashraf U, Papamargaritis D, Davies MJ. What is the pipeline for future medications for obesity? *Int J Obes (Lond)*. 2025;49(3):433-451. <https://doi.org/10.1038/s41366-024-01473-y>

109. Chopra V, Khan MS, Abdelhamid M, et al. iCARDIO Alliance global implementation guidelines on heart failure 2025. *Global Cardiol*. 2025;34(7):e55-e82. <https://doi.org/10.4081/cardio.2025.70>

KEY WORDS GLP-1 receptor agonists, heart failure, obesity, weight loss